

MySQL Replication: Local Windows 10 → Railway

Keep a warm online copy of the hospital database for failover when the on-premises server fails

SCENARIO

Primary HMS on Windows 10 + XAMPP MySQL; secondary HMS + MySQL on Railway for DR fallback

PRIMARY

Windows 10 · localhost MySQL

SECONDARY

Railway MySQL + Railway Node service

SCRIPT

`scripts/sync-mysql-to-railway.js`

RPO TARGET

15–60 minutes (scheduled sync interval)

Version

1.0 — June 2026

Failover type

Manual DNS / URL switch

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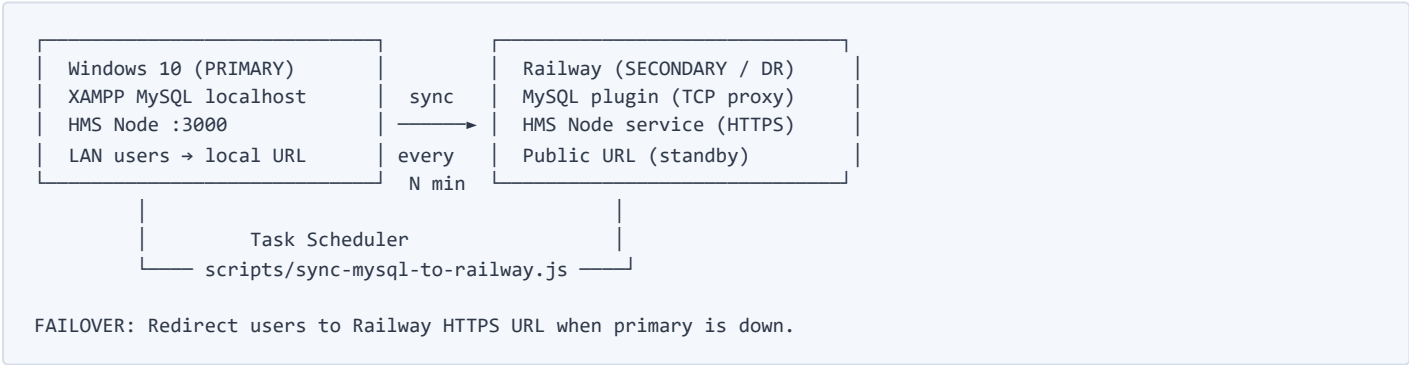
1. Strategy & terminology

Goal: If the Windows 10 hospital server fails (hardware, power, disk, OS), staff can continue on the Railway-hosted HMS using data synced from the local MySQL database.

Term	Meaning for this guide
Primary	Local Windows HMS + local MySQL (daily production)
Secondary / DR	Railway HMS + Railway MySQL (warm standby)
RPO	Recovery Point Objective — max data loss = time since last sync
RTO	Recovery Time Objective — time to switch users to Railway URL
Logical sync	<code>mysqldump</code> → import (recommended, included script)
Native replication	MySQL binlog streaming (advanced; Section 6)

Important: This is *warm standby* sync, not live active-active. Only one site should accept writes at a time to avoid split-brain conflicts.

2. Architecture



3. Railway MySQL setup

1 In your Railway project, click **New** → **Database** → **MySQL**.

- 2 Open the MySQL service → **Variables** tab. Note:
- `MYSQLHOST` (public proxy, e.g. `crossover.proxy.rlw.net`)
 - `MYSQLPORT`
 - `MYSQLUSER` / `MYSQLPASSWORD`
 - `MYSQLDATABASE` (often `railway`)

3 Ensure the MySQL plugin allows **public networking** (required for sync from hospital WAN).

- 4 Restrict access: use strong password; consider IP allowlisting if Railway plan supports it.

4. Railway HMS app (standby)

Deploy the main HMS application as a second Railway service in the same project (or separate project linked to same MySQL).

Variable	Value on Railway standby
DB_HOST	Railway MySQL host (MYSQLHOST)
DB_PORT	Railway MySQL port
DB_USER / DB_PASSWORD	From MySQL plugin
DB_NAME	railway or your database name
SESSION_SECRET	Same as local OR different (users re-login on failover)
LICENSE_SERVER_URL	Your license server HTTPS URL
HMS_PUBLIC_URL	Railway HMS public HTTPS URL (for push webhooks when on DR)

Generate Railway HTTPS domain for the HMS service. Keep this URL documented for failover.

During normal operations, train staff to use the **local** URL. Railway is the emergency URL only (or use DNS failover — Section 7).

5. Scheduled logical sync (recommended)

The repository includes `scripts/sync-mysql-to-railway.js` which:

1. Dumps local MySQL with `mysqldump` (XAMPP)
2. Connects to Railway MySQL over TLS TCP
3. Drops and re-imports tables (full refresh)
4. Verifies table count and sample row counts

5.1 Configure sync environment file

Create `C:\HMS_JS\docs\scripts\railway-sync.env` (do not commit secrets):

```
# Local XAMPP MySQL (primary)
LOCAL_DB_HOST=localhost
LOCAL_DB_PORT=3306
LOCAL_DB_USER=tssfsoas_Admin
LOCAL_DB_PASSWORD=your-local-password
LOCAL_DB_NAME=tssfsoas_HMS
XAMPP_MYSQL_BIN=C:\xampp\mysql\bin

# Railway MySQL (from plugin variables)
RAILWAY_MYSQL_HOST=crossover.proxy.rlwy.net
RAILWAY_MYSQL_PORT=36338
RAILWAY_MYSQL_USER=root
RAILWAY_MYSQL_PASSWORD=your-railway-password
```

```
RAILWAY_MYSQL_DATABASE=railway

# Optional dump path
SYNC_DUMP_PATH=C:\HMS-Backups\railway-sync\latest.sql
```

5.2 Manual test run

```
cd C:\HMS_JS
set RAILWAY_MYSQL_HOST=crossover.proxy.rlwy.net
set RAILWAY_MYSQL_PORT=36338
set RAILWAY_MYSQL_USER=root
set RAILWAY_MYSQL_PASSWORD=your-password
set RAILWAY_MYSQL_DATABASE=railway
node scripts/sync-mysql-to-railway.js
```

5.3 Automate with Windows Task Scheduler

1 Create folder `C:\HMS-Backups\railway-sync\logs` .

2 Create `C:\HMS_JS\docs\scripts\run-railway-sync.ps1` :

```
$envFile = "C:\HMS_JS\docs\scripts\railway-sync.env"
Get-Content $envFile | ForEach-Object {
    if ($_ -match '^(^#)=(.*)$') { Set-Item -Path "Env:$($matches[1].Trim())" -Value $matches[2].Trim()
    }
}
cd C:\HMS_JS
node scripts/sync-mysql-to-railway.js *>&1 | Tee-Object -FilePath "C:\HMS-Backups\railway-
sync\logs\sync-$(Get-Date -Format yyyyMMdd-HH:mm).log"
```

3 Task Scheduler → Create Task:

- Trigger: every **15 minutes** (or 30/60 based on RPO needs)
- Action: `powershell.exe -ExecutionPolicy Bypass -File C:\HMS_JS\docs\scripts\run-railway-sync.ps1`
- Run whether user is logged on or not · highest privileges
- Run only when network is available

4 Schedule during low activity if full dump is large; test duration first.

5.4 Sync window & production impact

- `mysqldump --single-transaction` avoids long table locks on InnoDB tables.
- Large databases (multi-GB) may take several minutes — adjust interval accordingly.
- During import, Railway DB is briefly inconsistent — run HMS on Railway in **maintenance mode** or keep it unused until sync completes.

6. Native MySQL replication (advanced)

For near-real-time RPO, MySQL binary log replication can stream changes from local master to Railway replica. This is more complex and may be limited by Railway MySQL privileges.

6.1 When to use

- Database > 2 GB and full dump every 15 min is too slow
- RPO requirement < 5 minutes
- IT staff comfortable with MySQL replication administration

6.2 Local master (XAMPP my.ini)

```
[mysqld]
server-id=1
log-bin=mysql-bin
binlog_format=ROW
expire_logs_days=7
bind-address=127.0.0.1
```

Opening MySQL to the public internet for replication is a security risk. Prefer VPN or Railway private networking if available. Never expose XAMPP MySQL to `0.0.0.0` without firewall IP restrictions.

6.3 Railway replica

Create replication user on master, take initial snapshot, configure Railway MySQL as replica with `CHANGE REPLICATION SOURCE TO` (MySQL 8). Verify `SHOW REPLICA STATUS\G` — `Replica_IO_Running: Yes`.

If Railway managed MySQL blocks replication commands, remain on Section 5 logical sync.

7. Failover procedure (local → Railway)

Stop writes on primary first. If primary is already dead, skip to step 3.

1 Confirm primary failure — server unreachable, MySQL down, or unrecoverable within RTO.

2 Run final sync if primary MySQL is still readable (from another machine or disk mount):

```
node scripts/sync-mysql-to-railway.js
```

3 Verify Railway database — connect with MySQL client; check `tbl_patient` row count vs last known good.

4 Start / verify Railway HMS — open `/__health` on Railway URL.

5 Redirect users:

- Update DNS A/CNAME for hospital hostname to Railway, OR
- Communicate emergency URL to all departments, OR
- Update reverse proxy / firewall rules

6 Update license server — ensure Railway HMS `HMS_PUBLIC_URL` matches active URL for push webhooks.

7 Post incident — document data loss window (time since last successful sync).

8. Fail-back to local Windows server

1 Repair or replace local hardware; restore OS and XAMPP from backup (see Backup DR guide).

2 Option A: Dump Railway MySQL → import to local (reverse sync — adapt script swapping LOCAL/RAILWAY).

3 Option B: Restore local from latest local backup then replay changes manually if gap is small.

4 Run full UAT on local before switching users back.

5 Re-enable scheduled sync local → Railway once local is primary again.

9. Monitoring & RPO/RTO

Metric	Target	How to monitor
Sync success	Every scheduled run	Task Scheduler history + log files in <code>C:\HMS-Backups\railway-sync\logs</code>
RPO	≤ sync interval (e.g. 15 min)	Timestamp in last log line
RTO	≤ 30 min staff communication	Quarterly failover drill
Railway HMS health	Weekly ping	<code>GET /__health</code> on standby URL

☐ Quarterly DR drill: run sync, fail over to Railway URL, test login + one clinical workflow

☐ Alert if sync log shows failure two consecutive runs

☐ Document Railway URL and credentials in sealed IT runbook

10. Troubleshooting

Problem	Solution
<code>ECONNREFUSED</code> to Railway	Check public networking on MySQL plugin; verify host/port/password
<code>mysqldump</code> not found	Set <code>XAMPP_MYSQL_BIN=C:\xampp\mysql\bin</code>
Import timeout	Increase Node memory; run sync off-peak; shrink dump (exclude logs if needed)
Schema errors on import	Script sanitizes MariaDB→MySQL8 defaults; ensure HMS versions match
Duplicate key after partial import	Script drops all tables first — check <code>clearRailwayDatabase</code> completed
Railway HMS shows old data	Confirm sync completed; restart Railway HMS service; check <code>DB_*</code> points to synced DB

